Boolean Algebra – Senior Division

1. **00-01 C3 Boolean Algebra**

   Simplify the expression (no XOR operator may appear in the answer):

   $$\overline{\overline{A+B}+\overline{AB}}+\overline{A\oplus B}$$

2. **01-02 C2 Boolean Algebra**

   Simplify completely: $(\overline{AB}C)(\overline{A}C)$

3. **01-02 C2 Boolean Algebra**

   List all ordered triples (A,B,C) that make the following expression TRUE

   $$AB\oplus\overline{C+(CB+A)}$$

4. **01-02 C3 Boolean Algebra**

   Simplify completely: $(A+B)\oplus AB$

5. **02-03 C2 Boolean Algebra**

   List all of the following that are NOT equivalent to:

   NOT (B AND (NOT A OR NOT B) OR A AND NOT (B AND NOT A OR NOT B))

   a. NOT ((NOT A AND B) OR A AND B AND NOT (B AND NOT A))
   b. NOT B
   c. NOT A AND B OR A AND B
   d. NOT (NOT A AND B OR A AND B)
   e. B

6. **02-03 C2 Boolean Algebra**

   Find all the ordered triples that make the following expression FALSE:

   A AND NOT B XOR B AND (NOT A OR C)

7. 02-03 C3 Boolean Algebra

Find all ordered triples that make the following expression TRUE:

$$\overline{A}B + B(A + \overline{C}) + (\overline{C} + B)(\overline{A} + \overline{B})$$

8. 03-04 C2 Boolean Algebra

Find all ordered pairs that make the following expression TRUE.

$$\overline{A}(AB + \overline{A})$$

9. 03-04 C2 Boolean Algebra

Simplify the following expression: $\overline{A(\overline{B} + C)} + \overline{A\overline{B} + C}$

10. 03-04 C3 Boolean Algebra

Simplify the following Boolean expression: $A(\overline{(A + \overline{B})(B + C)} + B + C)$

11. 04-05 C2 Boolean Algebra

Simplify the following expression.
The simplified solution must have just one NOT.

$$\overline{A + (B + \overline{C})} + \overline{B}(\overline{A + C})$$

12. 04-05 C2 Boolean Algebra

How many ordered triples make the following expression TRUE?

$$A(B \oplus C) + \overline{AB}$$

13. 04-05 C3 Boolean Algebra

Simplify the following expression: $\overline{\overline{\overline{A}B + C} + \overline{B}(A + \overline{C})}$

14. 05-06 C2 Boolean Algebra

Which ordered triples make the following expression FALSE?

$$\overline{\overline{A}(\overline{B}+\overline{C})}+(A\oplus\overline{BC})$$

15. 05-06 C2 Boolean Algebra

Simplify: $\overline{\overline{\overline{A\overline{B}(1+\overline{C})}+C(\overline{A}+B)}+\overline{A}(B\overline{C}+\overline{A})}$

16. 05-06 C3 Boolean Algebra

Simplify the following Boolean expression: $(A+B\overline{C})(\overline{A}\overline{B}+C)$

17. 06-07 C2 Boolean Algebra

Simplify completely: $(\overline{\overline{A}B+AC})(\overline{\overline{C}+B})A$

18. 06-07 C2 Boolean Algebra

How many ordered triples make the following expression TRUE?

$$(\overline{A}\oplus B)(\overline{B}\oplus C)(\overline{C}\oplus\overline{A})$$

19. 06-07 C3 Boolean Algebra

Simplify: $AB(C+A)+\overline{B}(A+BC)$

20. 07-08 C2 Boolean Algebra

Simplify completely: $\overline{\overline{A(\overline{B}+C)}+B(AC+B)}$

21. 07-08 C2 Boolean Algebra

How many ordered triples make the following expression TRUE?

$$A(B \oplus \overline{C}) + \overline{B}(AC \oplus \overline{B})$$

22. 07-08 C3 Boolean Algebra

Simplify: $(A + BC)(\overline{A}\,\overline{B} + C)(\overline{A} + \overline{B}C)(A\overline{B} + \overline{C})$

23. 08-09 C2 Boolean Algebra

Simplify the following Boolean expression using the fewest number of operators:

$$\overline{A}(B + A) + A\overline{B}$$

24. 08-09 C2 Boolean Algebra

How many ordered triples make the following expression TRUE?

$$A(B + \overline{C}) + \overline{BC} + \overline{A(BC + A)}$$

25. 08-09 C3 Boolean Algebra

List all ordered triples that make the following expression TRUE:

$$\overline{\overline{A\overline{B} + A\overline{C}} + (AB\overline{C} + A\overline{B}C)}$$

26. 09-10 C2 Boolean Algebra

Simplify the following Boolean expression: $(A + B)(B + \overline{A}C) + AB\overline{C} + B(\overline{A} + \overline{C})$

27. 09-10 C2 Boolean Algebra

List the ordered triples that make the following expression FALSE.

$$\overline{A\overline{B}+C}+A(B+\overline{C})\oplus \overline{A}BC$$

28. 09-10 C3 Boolean Algebra

How many ordered triples make the following FALSE?

$$\overline{(\overline{A+\overline{B}})(\overline{B+C})(\overline{A}B)}$$

29. 10-11 C2 Boolean Algebra

Which ordered triples make the following Boolean expression TRUE?

$$A(B+\overline{C})\oplus \overline{A}B$$

30. 10-11 C2 Boolean Algebra

Simplify the following Boolean expression: $A(\overline{A\overline{B}+BC})(\overline{A+B}+C)$

31. 10-11 C3 Boolean Algebra

Simplify the following expression: $\overline{(\overline{A}+B)(\overline{A}+B+\overline{C})+AB(\overline{A}+C)}$

32. 11-12 C3 Boolean Algebra

Simplify the following Boolean expression: $\overline{(A+B)+(\overline{B}+C)}+\overline{A}(\overline{B+C})$

33. 11-12 C3 Boolean Algebra

Which ordered triples make the following Boolean expression TRUE?

$$B(\overline{A(\overline{B}+C)})(\overline{C}(\overline{AB}))$$

34. 12-13 C3 Boolean Algebra

Simplify the following Boolean expression: $\overline{(\overline{A+\overline{B}})(AB)}+\overline{(A+B)(\overline{\overline{A}\,\overline{B}})}$

35. 12-13 C3 Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{A\overline{B}+C}+\overline{\overline{A}+C}+\overline{B\overline{C}}$$

36. 13-14 C3 Boolean Algebra

Which ordered triple(s) make the following Boolean expression TRUE?

$$(AB+\overline{C})(\overline{A}+BC)(A+\overline{B}+C)$$

37. 13-14 C3 Boolean Algebra

Simplify the following Boolean expression: $A\overline{B}(AB+\overline{C})+\overline{C}(A+\overline{B}C)$

38. 14-15 C3 Boolean Algebra

Simplify the following Boolean expression: $(\overline{\overline{A}+B})(B+C)(\overline{A+\overline{C}})(A\overline{B}+BC)$

39. 14-15 C3 Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{B}(\overline{A}C + B\overline{C}) + \overline{C}(AC + \overline{B}) + \overline{\overline{B} + C}$$

40. 15-16 C3 Boolean Algebra

Simplify the following Boolean expression: $\overline{A(B + \overline{C})} + \overline{B}\overline{C} + \overline{\overline{A} + B + \overline{C}}$

41. 15-16 C3 Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$\overline{AB}(A + \overline{B}C) + \overline{A}(\overline{BC} + A)$$